MARKETING MIX MODELING

Marketing Channel Performance & Model Analysis

PROJECT REPORT

A structured review of the Marketing Mix Modeling Power BI project, focusing on revenue trends, marketing investment, channel contribution and model-performance reporting.

01 / EXECUTIVE OVERVIEW

Executive Summary

Purpose, report coverage and key messages

This report documents a Marketing Mix Modeling (MMM) and marketing-channel performance project presented through a Power BI dashboard. The supplied report contains two main pages: an Overview page and a Model Performance page. Together, these pages organize information about sales revenue, marketing expenditure, channel-level performance, contribution and predicted revenue.

The dashboard is designed to help a business move beyond looking at total marketing spend alone. It brings investment and revenue measures into one analytical view so that decision-makers can compare channels, monitor changes over time and investigate how marketing activity relates to business outcomes.

What the dashboard contains

Dashboard area	Visible analytical content	Business question supported
Overview	Sales revenue, total spend, monthly revenue trend	How are revenue and marketing investment changing?
Channel performance	Steady-state revenue and marginal ROAS by channel	Which channels appear more productive?
Model performance	Actual versus predicted revenue by month	How closely do model predictions track observed revenue?
Contribution and ranking	Channel contribution share and effectiveness ranking	How is channel contribution represented and compared?
Spend distribution	Monthly spend by channel	How is marketing investment distributed?

Important interpretation note

Evidence-based reporting

The Power BI file structure identifies the dashboard visuals and the fields used in them. The underlying model's numeric results were not available for reliable extraction in this review. Therefore, this report explains the design, measures and appropriate interpretation without inventing revenue totals, rankings, ROAS values or model-accuracy scores.

Report structure

- Pages 3–4: project context, objectives, scope and analytical data structure.
- Pages 5–6: MMM concepts, performance measures and dashboard design.
- Pages 7–8: interpretation approach, recommendations and implementation considerations.
- Page 9: conclusion, limitations and next steps.

Introduction, Objectives & Scope

Why marketing mix analysis matters

1. Introduction

Organizations use multiple marketing channels to reach customers. These channels may include paid social media, search, display advertising, email, offline media and other promotional activities. Each channel can require a different level of investment and may contribute to revenue in a different way. Comparing them is difficult when spending and revenue information is reviewed separately.

Marketing Mix Modeling provides an analytical framework for estimating how marketing activities and other factors relate to a business outcome, commonly sales or revenue. A model can help quantify relationships, compare channel effectiveness and support more informed allocation of marketing budgets. The Power BI project translates selected model outputs into interactive visual summaries.

2. Project aim

Overall aim

To present marketing investment, revenue outcomes and model-derived channel performance measures in a clear Power BI report that supports monitoring and evidence-based marketing decisions.

3. Specific objectives

1. Present headline revenue and marketing-spend measures in a single overview.
2. Show the monthly movement of sales revenue to support trend monitoring.
3. Compare channels using steady-state revenue and marginal return on advertising spend (mROAS) fields.
4. Display actual and predicted revenue together to support model-performance review.
5. Summarize channel contribution share and effectiveness ranking.
6. Show the distribution of monthly marketing spend across channels.

4. Scope

Included in this report	Outside the verified scope
Review of the two Power BI pages and their visual fields	Independent validation of the underlying source dataset
Explanation of MMM concepts and dashboard measures	Re-estimation of the statistical model
Interpretation guidance and practical recommendations	Claims about causal impact or exact model accuracy
Suggested quality checks and future improvements	Verified numerical rankings where source values are unavailable

5. Intended users

The dashboard is most relevant to marketing managers, analysts, finance teams and business decision-makers who need a consolidated view of marketing investment and revenue-related performance. It can also support project evaluation by demonstrating how analytical outputs are communicated through business intelligence visuals.

Data Structure & Preparation

Understanding the information represented in the report

The visual definitions in the supplied PBIX reference several analytical tables. The names indicate that the project separates channel response information, channel effectiveness summaries and holdout predictions. This structure is consistent with a workflow in which model outputs are prepared in analytical tables and then explored in Power BI.

Referenced table	Fields visible in dashboard	Role in the report
channel_response_curves	channel, month, monthly_spend_lkr, steady_state_revenue_lkr, marginal_roas	Channel-level spend, response and return measures.
channel_effectiveness_summary	channel, contribution_share_pct, effectiveness_rank	Summary measures for channel contribution and relative effectiveness.
holdout_predictions	month, sales_revenue_lkr, predicted_revenue_lkr	Comparison of observed and predicted revenue for a holdout period.

Recommended data preparation workflow

Step	Activity	Why it matters
1. Source review	Identify source files, fields, units and time coverage.	Establishes what each field represents.
2. Data cleaning	Check missing values, duplicates, invalid dates and inconsistent channel labels.	Reduces avoidable reporting errors.
3. Standardization	Align date formats, channel names and currency units.	Makes records comparable across tables.
4. Model-output checks	Confirm predicted values, response metrics and summary fields are valid.	Protects the interpretation of model outputs.
5. Power BI integration	Load prepared tables and verify relationships and aggregation behavior.	Ensures visuals summarize the intended data.

Data-quality checks to complete

- Confirm that all revenue and spend fields use the same currency convention (LKR is indicated in field names).
- Check that month values are consistent and sort chronologically rather than alphabetically.
- Ensure channel names match across response, effectiveness and prediction-related tables.
- Verify that contribution-share percentages use a consistent denominator and sum appropriately for the selected scope.
- Check that the ranking field is treated as a rank, not automatically summed as if it were a quantity.
- Confirm that the actual-versus-predicted comparison uses the intended holdout period.

Model governance
Document the source, refresh date, transformations and definitions for each key measure. This makes the dashboard easier to audit, maintain and explain.

Marketing Mix Modeling & Key Measures

How the analytical measures should be understood

1. What is Marketing Mix Modeling?

Marketing Mix Modeling is a statistical approach that examines how different marketing activities and other business factors relate to an outcome such as sales or revenue. A typical model uses historical observations to estimate relationships between marketing inputs and the business outcome. The results can be used to compare channel contributions and explore potential budget scenarios.

2. Core measures represented in the PBIX

Measure	Plain-language meaning	Interpretation guidance
Sales revenue (LKR)	Observed revenue recorded for a period.	A business outcome; interpret alongside time period and context.
Total spend (LKR)	Marketing expenditure represented in the report.	Shows the level of investment, not effectiveness by itself.
Monthly spend (LKR)	Spend associated with a channel and month.	Useful for identifying changes in allocation.
Steady-state revenue (LKR)	A model-related estimate of longer-run or stabilized revenue response.	Use the project's precise model definition before interpreting.
Marginal ROAS	Estimated incremental revenue return associated with additional advertising spend.	Higher values may indicate stronger marginal efficiency, subject to model assumptions.
Contribution share (%)	A channel's share of the modeled contribution total.	Check the denominator and whether the shares total 100%.
Effectiveness rank	Relative position assigned to a channel.	Confirm whether rank 1 means best and how ties are handled.
Predicted revenue (LKR)	Revenue estimated by the model for a period.	Compare with actual revenue using appropriate error measures.

3. ROAS and marginal ROAS

Return on Advertising Spend (ROAS) is commonly expressed as revenue attributed to advertising divided by advertising cost. Marginal ROAS focuses on the estimated return from an additional unit of spend, rather than the average return across all spending. These measures answer different questions and should not be treated as interchangeable.

Interpret with care

A channel with a high contribution share is not automatically the most efficient channel. Contribution describes relative modeled contribution, while marginal ROAS focuses on the estimated return from additional spend.

4. Actual versus predicted revenue

A line chart comparing actual and predicted revenue helps reveal whether the model follows the broad pattern of observed results. Close alignment is encouraging, but visual similarity alone is not a complete validation. The model should also be assessed using numerical error measures and a clearly defined holdout period.

Dashboard Design & Visual Analysis

Review of the two report pages in the supplied PBIX

1. Overview page

The Overview page brings together headline indicators and channel-level comparisons. The visual definitions show a sales-revenue card, a total-spend card, a monthly sales-revenue line chart, and channel comparisons for steady-state revenue and marginal ROAS. This arrangement gives users a quick summary before they explore more detailed channel measures.

Visual	Fields used	Purpose
Revenue KPI card	sales_revenue_lkr	Highlights the revenue measure selected by the report's aggregation.
Spend KPI card	total_spend_lkr	Shows the marketing investment measure.
Monthly revenue trend	month; sales_revenue_lkr	Displays how revenue changes over time.
Channel revenue comparison	channel; steady_state_revenue_lkr	Compares modeled steady-state revenue by channel.
Marginal ROAS comparison	channel; marginal_roas	Compares estimated marginal return across channels.

2. Model Performance page

The Model Performance page focuses on model outputs and channel summaries. It includes an actual-versus-predicted revenue trend, a channel contribution pie chart, a channel performance ranking table and a monthly-spend comparison by channel.

Visual	Fields used	Purpose
Actual vs predicted line chart	month; sales_revenue_lkr; predicted_revenue_lkr	Reviews the fit of predicted revenue against observed revenue.
Contribution pie chart	channel; contribution_share_pct	Shows the relative contribution share by channel.
Performance ranking table	channel; marginal_roas; effectiveness_rank	Supports side-by-side channel review.
Monthly spend bar chart	channel; monthly_spend_lkr	Compares spend levels across channels.

3. Design strengths

- Two-page structure separates business overview from model-focused analysis.
- Line charts are suitable for examining monthly patterns.
- Bar charts support channel comparisons across numerical measures.
- Cards help users identify headline indicators quickly.
- Tables can present detailed channel measures where exact values matter.

Suggested usability improvement

Add clear date filters, channel slicers and concise measure definitions. Use consistent currency formatting and ensure chart titles describe the period and aggregation being shown.

06 / INTERPRETATION

How to Read the Results

A disciplined approach to drawing conclusions

The dashboard is intended to support analysis rather than replace business judgment. The following approach can be used when reviewing the live report and confirming the actual values shown by the visuals.

1. Start with the overall picture

Review total revenue and total spend together. Consider whether revenue and investment move in similar or different directions over the selected period. A change in revenue should not automatically be attributed to marketing without considering seasonality, pricing, promotions, distribution, competition and other influences.

2. Review the monthly trend

Use the monthly revenue chart to identify sustained increases, declines, recurring seasonal patterns or unusual movements. Check whether the period contains incomplete months or exceptional events that may distort comparisons.

3. Compare channels using more than one measure

Question	Measure to inspect	What to avoid
Which channel contributes more?	Contribution share (%)	Do not assume contribution share equals efficiency.
Which channel may be more efficient at the margin?	Marginal ROAS	Do not treat a model estimate as guaranteed future return.
Where is budget concentrated?	Monthly spend (LKR)	High spend does not necessarily mean high performance.
Which channel is ranked higher?	Effectiveness rank	Confirm rank direction and ranking methodology.

4. Assess model behavior

Compare the actual and predicted revenue lines over the holdout period. Look for periods where the lines diverge, whether the model consistently over-predicts or under-predicts, and whether large errors occur around peaks or troughs. A model may track the general trend but still make material errors in particular months.

5. Validate before communicating a finding

- Check the selected date range, filters and channel selection.
- Read the axis labels, units and aggregation method.
- Confirm that the value is actual, predicted, modeled or observed.
- Compare at least two relevant measures before making a channel recommendation.
- Record the dashboard filter context when capturing a result for a report or presentation.

Reporting discipline
A strong insight states what changed, where it changed, how large the change was, and what evidence supports the explanation. Avoid writing a conclusion until the corresponding dashboard values have been verified.

Business Recommendations & Next Steps

Turning dashboard analysis into responsible action

1. Recommendations for marketing decision-makers

7. Use marginal ROAS alongside contribution share when reviewing channel efficiency. This avoids prioritizing a channel based on size alone.
8. Review spend and revenue trends together at a consistent monthly level. Investigate sudden changes before adjusting budgets.
9. Treat model-based channel rankings as decision support. Combine them with business context, operational capacity and campaign objectives.
10. Use controlled budget experiments where practical. Small, measured changes can help test whether modeled responses are reflected in real outcomes.
11. Review channel performance regularly and document the reasons for budget changes so that future results can be evaluated.

2. Recommendations for improving the Power BI report

Improvement	Practical implementation	Expected benefit
Date filtering	Add a shared month/date slicer across both pages.	Consistent time-based analysis.
Channel filtering	Add a channel slicer with synchronized behavior.	Faster investigation of individual channels.
Measure definitions	Add tooltips or an information panel for ROAS, contribution and ranking.	Reduces ambiguity for users.
Model validation KPIs	Display MAE, RMSE and MAPE where appropriate and valid.	Adds quantitative evidence beyond line-chart comparison.
Data freshness	Show last refresh date and reporting period.	Improves transparency.
Ranking logic	Use a clearly defined rank measure and appropriate aggregation.	Prevents misleading totals or sort order.

3. Model validation and monitoring

The model should be reviewed on data that was not used to fit it. Suitable measures may include Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE), provided the data and business context make each measure appropriate. MAPE can be unstable when actual values are zero or close to zero. Validation should also check whether errors are systematically higher for particular channels, months or business conditions.

4. Suggested operating rhythm

Frequency	Suggested activity
Monthly	Refresh the dashboard, check data quality and review revenue/spend movements.
Quarterly	Review channel contribution, marginal ROAS and model prediction errors.
After major campaigns	Compare expected and observed outcomes and document unusual factors.
At model refresh	Reassess assumptions, feature definitions, validation results and model stability.

08 / CLOSING SECTION

Conclusion, Limitations & References

Final assessment and recommended follow-up

Conclusion

The supplied Power BI project provides a structured view of marketing channel performance through two complementary report pages. The Overview page focuses on revenue, marketing spend, monthly trends, steady-state revenue and marginal ROAS. The Model Performance page adds actual-versus-predicted revenue, channel contribution share, effectiveness ranking and monthly channel spend.

This design creates a useful foundation for reviewing marketing investment and model outputs in one place. Its strongest value is the ability to bring multiple measures together, helping users ask better questions about channel contribution, marginal efficiency and the behavior of revenue predictions.

To make the report more decision-ready, the next stage should verify the underlying values, document each metric, add quantitative model-validation measures and ensure that filters and aggregations behave consistently. Business recommendations should be based on verified dashboard results and interpreted alongside factors beyond marketing.

Limitations of this document

- This report was prepared from the structure and visual definitions available in the supplied PBIX file.
- The underlying numeric model outputs were not independently extracted or validated for this document; therefore, no exact performance values or channel winners are claimed.
- The exact statistical model specification, source-data period, training procedure and model assumptions should be documented from the project workflow or assignment materials.
- Marketing Mix Modeling estimates relationships under model assumptions; results should not be interpreted as proof of causation without appropriate design and validation.

References and project materials

- User-supplied Power BI project: MMM_PROJECT_dilshan.pbix (dashboard pages: Overview and Model Performance).
- Microsoft Power BI documentation: https://learn.microsoft.com/power-bi/
- Google Meridian – Marketing Mix Modeling resources: https://developers.google.com/meridian
- Robyn – open-source Marketing Mix Modeling project: https://facebookexperimental.github.io/Robyn/

Recommended final check before submission

Open the PBIX in Power BI Desktop and insert verified KPI values, model-validation results, source-data details and dashboard screenshots if your assignment rubric requires evidence. Replace any generic project metadata with the exact details from your brief.

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